

ClinicOS

Version: 1.0

System type: B2B clinical management SaaS

Model: Multi-clinic

Target audience: Small, modern clinics (2–10 users)

1. Introduction

1.1 Objective

This document outlines the functional and non-functional requirements for ClinicOS.

Functional requirements represent what the system must do.

Non-functional requirements represent how the system must operate, including quality, security, performance, and user experience.

2. Initial system scope

The first version of ClinicOS will focus on:

- ClinicOS V1:
 - Authentication
 - Clinic management
 - Users and permissions
 - Patients
 - Doctors
 - Schedule
 - Appointments
 - Dashboard
 - Basic settings

Future features:

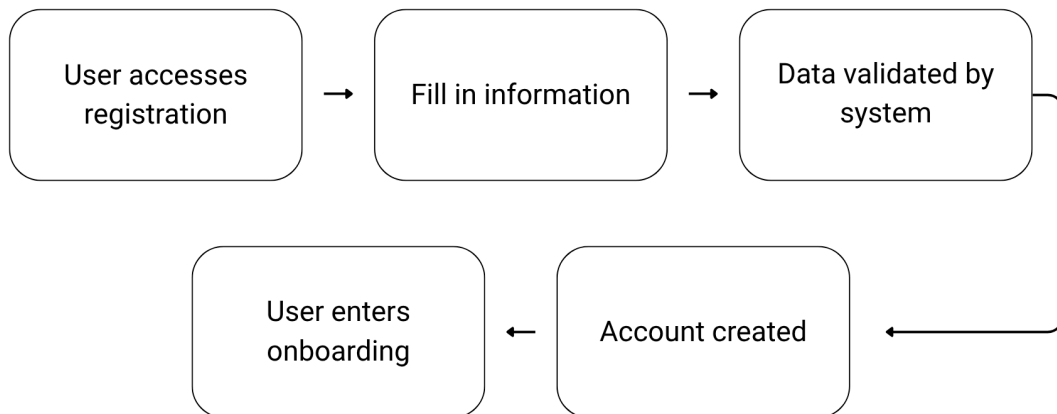
- V2
 - AI
 - Automations
 - WhatsApp
 - Advanced reports
 - Financials
 - Integrations
- V3
 - Mobile app
 - Advanced multi-unit capabilities
 - Marketplace
 - Advanced AI

3. Functional Requirements

Module 01 — Authentication

- **RF001 — Account registration**
 - **Description:** The system must allow a new client to create an account on ClinicOS.
 - **Primary actor:** Account owner.
 - **Required data:**

- Name;
- Email;
- Password;
- Organization name.
- **Basic flow:**



- **RF002 – Login**
 - **Description:** The system must allow registered users to authenticate.
 - **Input:**
 - Email;
 - Password.
 - **Output:** Authenticated user.
- **RF003 — Password recovery**
 - The system must allow access recovery via the registered email.
- **RF004 — Session management**
 - The system must keep users authenticated during a valid session.

Module 02 — Clinic Management

- **RF005 — Create clinic**
 - The owner must be able to create a new unit.
 - **Data:**
 - Name;
 - CNPJ (Tax ID);
 - Phone;
 - Address;
 - Operating hours.
- **RF006 — Manage clinics**

- **The system must allow:**
 - viewing clinics;
 - editing information;
 - activating/deactivating units.
- **RF007 — Data separation by clinic**
 - The system must ensure that users have access only to data from authorized clinics.
 - **Example:**
 - **Brasília Clinic:**
 - Patient A
 - Patient B
 - **Goiânia Clinic:**
 - Patient C
 - Patient D
 - Users of the Brasília Clinic cannot view data from the Goiânia Clinic.

Module 03 — Users and Permissions

- **RF008 — Create users**
 - The administrator can register:
 - doctors;
 - receptionists;
 - administrators.
- **RF009 — Define user role**
 - Each user must have a role:
 - ADMINISTRATOR
 - RECEPTIONIST
 - DOCTOR
- **RF010 — Edit user**
 - Allow changes to:
 - name;
 - email;
 - role/position;
 - status.
- **RF011 — Deactivate user**
 - Users can be blocked without permanent deletion.

Module 04 — Patients

- **RF012 — Register patient**
 - The system must allow the creation of patient records.
 - Initial data:
 - Name
 - CPF (Tax ID)
 - Date of birth
 - Phone

- Email
 - Address
 - Notes
- **RF013 — List patients**
 - Allow:
 - search;
 - filters;
 - pagination.
- **RF014 — View patient profile**
 - The profile must display:
 - **Patient:**
 - Personal details
 - Appointments
 - History
 - Notes
- **RF015 — Edit patient**
 - Authorized users can update information.
- **RF016 — Delete patient**
 - The system must allow deletion while adhering to security rules.
 - Preferably:
 - Soft delete.
 - **Example:** active = false

Module 05 — Doctors

- **RF017 — Register doctor**
 - Allow registration of:
 - Name;
 - Specialty;
 - Professional license number;
 - Contact details.
- **RF018 — Manage doctors**
 - Allow:
 - listing;
 - editing;
 - removing;
 - viewing profile.

Module 06 — Schedule

- **RF019 — View schedule**
 - The system must display:
 - daily view;
 - weekly view;
 - filters by doctor.

- **RF020 — Create appointment**
 - Allow appointment creation.
 - **Data:**
 - Patient
 - Doctor
 - Date
 - Time
 - Notes
 - Status
- **RF021 — Edit appointment**
 - Allow changes to:
 - time;
 - doctor;
 - patient;
 - status.
- **RF022 — Cancel appointment**
 - Change appointment status.
 - **States:**
 - SCHEDULED
 - CONFIRMED
 - CANCELLED
 - DONE
 - NO_SHOW
- **RF023 — Prevent conflicts**
 - The system must prevent:
 - a doctor having two appointments at the same time;
 - invalid time slots.

Module 07 — Dashboard

- **RF024 — Administrative dashboard**
 - Show:
 - appointments for the day;
 - registered patients;
 - active doctors;
 - upcoming appointments.
- **RF025 — Personalized dashboard**
 - Content must vary based on the user.
 - **Example:**
 - **Administrator:** Metrics
 - **Reception:** Schedule
 - **Doctor:** My schedule

Module 08 — Settings

- **RF026 — User profile**
 - Allow:
 - changing name;
 - changing password;
 - preferences.
- **RF027 — Clinic configuration**
 - Allow:
 - editing details;
 - operating hours;
 - public information.

4. Non-Functional Requirements

Security

- **RNF001 — Secure authentication**
 - The system must store passwords using secure encryption.
 - **Example:** bcrypt / argon2
- **RNF002 — Authorization control**
 - Users may only perform actions permitted by their role.
 - **Example:**
 - Receptionist
 - cannot
 - change clinic settings
- **RNF003 — Data protection**
 - The system must protect sensitive patient information.
- **RNF004 — Auditing**
 - The system must log important actions.
 - **Example:**
 - **User:** Mariana
 - **Action:** Edited patient
 - **Date:** August 1, 2026

Performance

- **RNF005 — Response time**
 - Common operations must respond quickly.
 - **Target:** < 2 seconds
- **RNF006 — Scalability**
 - The system must support growth in:
 - clinics;
 - users;
 - patients;
 - appointments.

Availability

- **RNF007 — System online**
 - The SaaS must be continuously available.
 - **Target:** 99% uptime

Usability

- **RNF008 — Intuitive interface**
 - Users must be able to perform common tasks without training.
- **RNF009 — Responsiveness**
 - The system must work on:
 - Desktop
 - Tablet
 - Mobile

Compatibility

- **RNF010 — Browsers**
 - Support:
 - Chrome;
 - Edge;
 - Firefox;
 - Safari.

Maintenance

- **RNF011 — Organized code**
 - The project must follow:
 - modular architecture;
 - coding standards;
 - documentation.
- **RNF012 — Testability**
 - The system must allow for:
 - unit tests;
 - integration tests;
 - interface tests.

Privacy and LGPD

- **RNF013 — Consent**
 - The system must allow control over personal data.
- **RNF014 — Data deletion**
 - Users must be able to request the removal of information in accordance with LGPD rules.

5. Summary of requirements

- **FUNCTIONAL REQUIREMENTS**
 - WHAT DOES IT DO?
 - Login

- Patients
- Doctors
- Schedule
- Appointments
- Dashboard

- **NON-FUNCTIONAL REQUIREMENTS**

- HOW DOES IT WORK?
 - Secure
 - Fast
 - Scalable
 - Responsive
 - User-friendly
 - LGPD-compliant

6. Requirement priority

- **High priority (V1)**

- RF001
- RF002
- RF005
- RF008
- RF012
- RF017
- RF019
- RF020
- RF024

- **Medium priority**

- RF014
- RF021
- RF022
- RF026
- RF027

- **Low priority / future**

- AI
- Financial
- Reports
- Integrations
- Advanced automations

Conclusion

With this document, ClinicOS moves beyond being merely an idea and takes on an initial software design.